distance between the false ceiling and the ceiling slab is large, it may be more cost-effective to include the lead in the false ceiling, because this will provide additional protection to both the floor above and the adjacent rooms; in this case, Code 3 lead will almost certainly suffice.

The control cubicle for a CT scanner would normally include a protected door. It is unlikely that a cubicle without a door would be acceptable, given that a number of staff, including radiographers, radiologists, nurses and other clinicians may need to be accommodated. Moreover, tertiary scatter from the adjacent wall into the area occupied by staff is likely to be significant. The level of tertiary scatter from the wall in μGy can be calculated using Equation 2.6 with the DLP workloads:

$$S_{wall} = 12 \times (DLP_b + DLP_h/2) \times e^{-0.56d} \quad (8.5)$$

where d is the distance of the isocentre from the wall adjacent to the door in metres.

8.5 Worked examples

A CT scanner in a busy city hospital is to be located in a room measuring 5.55 × 7 m, with the operator's cubicle in an adjacent room. The separation of floor and ceiling slabs is 4.0 m. Each is constructed from lightweight concrete (density, 1840 kg m^{-3}) with a minimum thickness of 100 mm. The scanner isocentre is located 0.9 m above floor level. The scanner is to be located towards the left side of the room, as shown in the plan (see Figure 8.5). Patients will enter through the double door on the right-hand wall. The scanner is positioned at an angle of 12° so that the operator can obtain

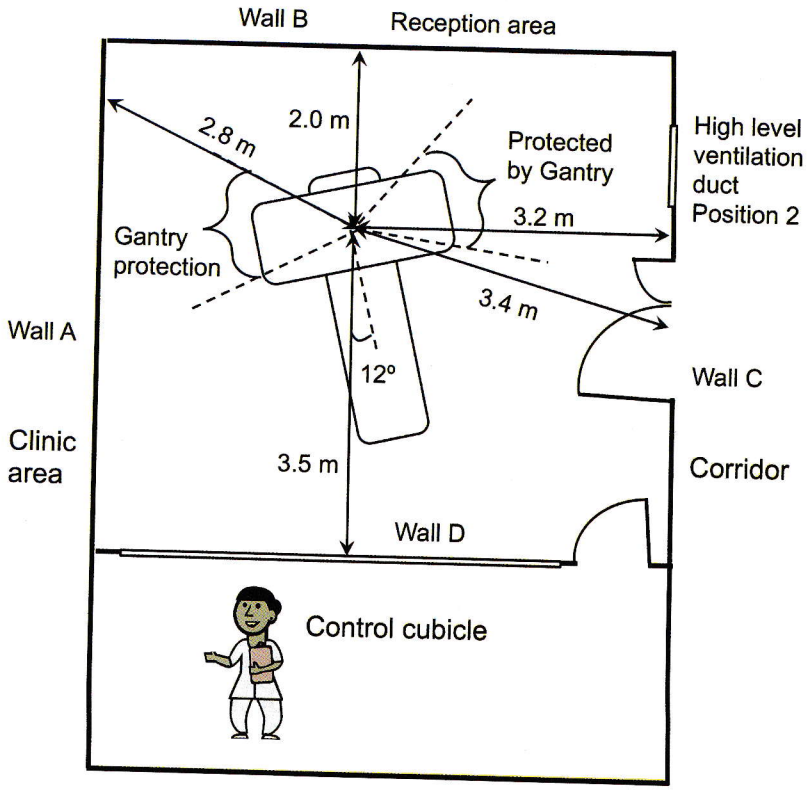


Figure 8.5 Plan of the CT room layout used in the worked example. Distances to the respective walls A–D) that will be used in the calculations of unattenuated scatter d_A, d_B, d_C, and d_D are given.

a good view of the patient within the gantry. The door from the cubicle into the room is located in the right-hand corner to maximise the space, and the operator cubicle window extends for most of the remaining length of the wall to provide a complete view of the scanner and the other door. The distance to Wall A on the left side of the figure is 2.5 m, that to Wall B is 2 m, that to Wall C containing the door giving access to the corridor is 3.2 m and that to the control cubicle window is 3.5 m.

The projected workload for the CT scanner comprises 90 body and 60 head examinations per week; few are undertaken at 140 kV, so transmission data for 120 kV will be used. Average DLPs for both the body and the head are taken as 900 mGy cm, which gives a workload that is towards the busier end of the range. Substituting the weekly workload for body and head scans into Equation 8.2 together with the CT scatter factors (see Table 8.1) gives an annual DLP for the body:

$$DLP_b = 90 \times 50 \times 900\ \text{mGy cm} = 4.05 \times 10^6\ \text{mGy cm} = 4.05 \times 10^3\ \text{Gy cm} \quad (8.6)$$

and for the head:

$$DLP_h = 60 \times 50 \times 900\ \text{mGy cm} = 2.70 \times 10^6\ \text{mGy cm} = 2.70 \times 10^3\ \text{Gy cm} \quad (8.7)$$

The air kerma at 1 m from the isocentre (K_{CT1}) to the front of the gantry then equals:

$$K_{CT1} = (4.05 \times 10^6 \times 0.36) + (2.70 \times 10^6 \times 0.14)\ \mu\text{Gy} = 184 \times 10^4\ \mu\text{Gy} \quad (8.8)$$

A slightly lower value can be used for the rear, if this offers any advantage. The air kerma at 1 m from the isocentre (K_{CT2}) at the rear of the couch is equal to:

$$K_{CT2} = (4.05 \times 10^6 \times 0.3) + (2.70 \times 10^6 \times 0.14)\ \mu\text{Gy} = 160 \times 10^4\ \mu\text{Gy} \quad (8.9)$$

The gantry offers protection at angles between −30° (front of gantry) and +40° (rear) to the plane of rotation (see Figure 8.5). The protection offered by the gantry would give an equivalent air kerma at 1 m from the isocentre (K_{gantry}) equal to:

$$K_{gantry} = (4.05 \times 10^6 \times 0.04) + (2.70 \times 10^6 \times 0.014)\ \mu\text{Gy} = 20 \times 10^4\ \mu\text{Gy} \quad (8.10)$$

Calculations of shielding requirements for various aspects of the scanner room are given in the Examples 8a–e using a dose constraint of 0.3 mGy per annum. Results are summarised in Table 8.3.

Example 8a: walls to the rear of the gantry

Wall A

Scatter air kerma at the rear side of the gantry is incident on Wall A at angles between 40° and 90° with the scan plane. Because the gantry is aligned at an angle of 12°, the nearest point of exposure will be at an angle of (40° − 12°) = 28°. The distance to the wall at the rear from the isocentre is $d_A = 2.5/\cos 28° = 2.8$ m. At the front, the higher scatter is incident at angles over 30° to the scan plane, so here the highest scatter will be at an angle of (30° + 12°) = 42°. Because the difference in the cosines of the two angles of incidence is approximately 1.2, similar to the difference in scatter factors on the front and rear (Table 8.1), the calculations for either side of the gantry will give a similar result. The occupancy in the clinic on the other side of the wall is 50%.

Box 8.1 **Calculation for Wall A**

The annual air kerma incident on the wall is given by:

$K_{incA} = K_{CTZ}/d_A^2 = 160 \times 10^4/2.8^2\ \mu Gy = 204 \times 10^3\ \mu Gy$

The required transmission for the barrier $= 300/(204 \times 10^3 \times 0.5) = 3.0 \times 10^{-3}$

The protection can be provided by 1.6 mm of lead or 140 mm of standard concrete.

Wall B

Scatter is incident perpendicularly on Wall B at the rear of the gantry. The occupancy in the reception area on the other side is 100%.

Box 8.2 **Calculation for Wall B**

Scatter air kerma incident perpendicularly on Wall B is given by:

$K_{incB} = K_{CTZ}/d_B^2 = 160 \times 10^4/2.0^2\ \mu Gy = 400 \times 10^3\ \mu Gy$

The required transmission for the barrier $= 300/(400 \times 10^3 \times 1.0) = 0.75 \times 10^{-3}$.

The protection can be provided by 2.2 mm of lead or 176 mm of standard concrete.

Because this is the closest wall to the isocentre and is not protected by the gantry, this will determine the shielding of the walls for the whole room, which will be specified as Code 5 lead.

Example 8b: Wall C

Scatter air kerma at the front of the gantry is incident on Wall C at angles between 0° and 50° with the scan plane. The region where a higher scatter level is incident lies between 30° and 50° to the scan plane and includes the door to the corridor. The occupancy in the area outside the door is estimated to be 50%. The highest scatter kerma is incident at an angle of 30°−12° = 18° to the wall/door. The shortest distance (d_C) from the isocentre to scatter incident on the wall and door is:

$$d_C = 3.2/\cos 18^\circ = 3.4\ m \qquad (8.11)$$

The occupancy in the corridor outside the door is 20%.

Box 8.3 **Calculation for door in Wall C**

The annual air kerma incident on the door to the corridor (K_{CT1}) is given by:

$K_{incC} = K_{CT1}/d_C^2 = 184 \times 10^4/3.4^2\ \mu Gy = 159 \times 10^3\ \mu Gy$

The required transmission for the barrier $= 300/(159 \times 10^3 \times 0.2) = 9 \times 10^{-3}$.

The required transmission is provided by 1.2 mm of lead (Code 3).

The obliquity of the incident radiation could be taken into account in determining the lead thickness, but the advantages are limited for angles of incidence less than 30°.

Tertiary scatter from the adjacent wall could give staff behind the barrier a substantial dose if there were no door in place. The annual dose from tertiary scatter calculated using Equation 8.5 would be:

$$S_{ter} = 10 \times (DLP_b + DLP_h/2) \times e^{-0.5 \times 3.2} = 10 \times 5.4 \times 10^3 \times e^{-1.6}\ \mu Gy = 11 \times 10^3\ \mu Gy \qquad (8.12)$$

Thus, the door to a CT control cubicle is always likely to require shielding to protect staff behind the barrier from tertiary scatter.

Example 8c: CT operator's cubicle—Wall D

The occupancy in the area is taken to be 100%.

Box 8.4 Calculation for cubicle window

The annual air kerma incident on the window of the operator cubicle (K_{incD}), which is a distance d_D of 3.5 m from the isocentre is given by:

$K_{incD} = K_{CT1}/d_D^2 = 184 \times 10^4/3.5^2\ \mu Gy = 150 \times 10^3\ \mu Gy$

The required transmission for the barrier $= 300/(150 \times 10^3 \times 1.0) = 2.0 \times 10^{-3}$.

This transmission is provided by the equivalent of 1.8 mm of lead.

Wall above cubicle window

The cubicle window extends to a height of 2 m, but X-rays will be scattered down from the ceiling slab into the cubicle. The level of scatter for a ceiling of height 4 m with a 2 m high barrier can be predicted by substituting the annual DLP workload into Equation 8.4 with coefficients from Table 2.4 for a 4 m high ceiling and a barrier height b of 2 m.

Box 8.5 Calculation for wall above cubicle window

The air kerma scattered (S_{sec}) from the ceiling slab incident on an operator is given by:

$S_{sec} = [1.19 - (0.296 \times 2.0)] \times (4.05 + 1.35) \times 10^3\ \mu Gy = 3.2 \times 10^3\ \mu Gy$

The required transmission for the barrier $= 300/(3.2 \times 10^3 \times 1.0) = 0.09$.

Protection is provided by 0.5 mm of lead or 50 mm of standard concrete at 120 kV.

In this situation, even if the barrier was raised to a height of 3 m, the air kerma scattered from the ceiling slab would still be 1.6 mGy and additional protection would be required. The only situation where it may be possible to avoid protection to the roof is in departments with workloads at the lower end of the workload range (see Table 8.2), where protection up to the height of the false ceiling may be an option, but care must be taken in the specification because it would be considerably more expensive to install such shielding at a later stage.

Example 8d: ceiling—CT room

The radiosensitive organs of a person sitting in the room above the CT scanner will be about 0.5 m from the structural ceiling and so will be at a height of 3.6 m above the isocentre. The Radiology Department has no control (or knowledge) of any future changes in use of this area, so an occupancy of 100% is assumed.

Box 8.6 Calculation for ceiling

The distance along the line of greatest scatter will be $d_{ce} = (3.6/\cos 30°) = 4.2$ m.

Applying an inverse square law correction along the line of greatest scatter will give:

Annual air kerma incident on a person on the floor above (K_{incE}) is given by:

$K_{incE} = K_{CT1}/d_{ce}^2\ \mu Gy = 184 \times 10^4/4.2^2\ \mu Gy = 104 \times 10^3\ \mu Gy$

The required transmission for the barrier $= 300/(104 \times 10^3 \times 1.0) = 2.9 \times 10^{-3}$

The protection required in the roof is 1.7 mm of lead or 141 mm of standard concrete. The equivalent thickness (T_L) of lightweight concrete (*d*ensity, 1840 kg m^{-3}) would be:.

$T_L = 141 \times 2350/1840 = 180$ mm

The thickness of the roof slab is 100 mm and the distance along the line of scatter is 100/cos 30° = 112 mm. Because radiation is incident obliquely, an equivalent barrier thickness equal to the mean of the actual thickness and that in the direction of scatter can be assumed (106 mm). Thus the protection provided by the structural concrete in the floor is insufficient.

The thickness of barrier (T_L) required to attenuate radiation incident at angle θ can be calculated from the Equation 8.3. Thus for this example, $T_L = 180 \times (1 + \cos 30°)/2 = 168$ mm, so 68 mm of lightweight concrete would be required in addition to the existing 100 mm.

The existing lightweight concrete roof slab has a transmission of 13×10^{-3} at a 30° angle of incidence, so the addition of Code 3 lead will be more than sufficient to provide the additional protection.

Box 8.7 Specification of additional protection for the ceiling

A total thickness of 168 mm of lightweight concrete is required, or the addition of Code 3 lead to the existing 100 mm thick ceiling slab of lightweight concrete is needed to give the required level of protection for the floor above.

A similar level of protection will be required for the floor.

Example 8e: ventilation duct

A ventilation duct with an area of 0.5 m^2 must be sited in the wall adjacent to the corridor. The floor above contains a ward for which the occupancy factor is 100%.

Position 1 above door

It has been proposed that the duct would be located above the door and the diagonal distance through the penetration to a point 0.5 m above the next floor level (d_V) is 5.5 m.

Position 2 shielded by gantry

An alternative position would be to site the ventilation duct in a region protected by the gantry (see Figure 8.5).

Box 8.8 Calculation for ventilation duct protection—Position 1

$K_{vent1} = K_{CT1}/d_V^2 = 184 \times 10^4/5.5^2\ \mu Gy = 61 \times 10^3\ \mu Gy$

The required transmission for protection = $300/(61 \times 10^3 \times 1.0) = 5.0 \times 10^{-3}$.

This transmission is given by 161 mm of lightweight concrete for direct incidence. As the angle of incidence is 30°, this equates to:

$T_L = 161 \times (1 + \cos 30°)/2 = 150$ mm

Therefore the 100 mm thick ceiling slab will not provide sufficient protection. A baffle made from Code 3 lead would be required to protect against direct exposure.

8.6 Things to be aware of

The ceiling slab may not provide sufficient protection for rooms on the floor above to the side of the scanning room should the wall protection not extend up to the underside of the soffit. Areas on all sides of the scanning room are likely to require protection against tertiary scatter from the ceiling slab. The protection requirement in walls above 2 m is unlikely to exceed that provided by Code 3 lead.

Box 8.9 Calculation for ventilation duct protection—Position 2 shielded by gantry

If the ventilation duct were to be sited in a region protected by the gantry (see Figures 8.4 and 8.5), the air kerma level would be:

$K_{vent2} = K_{gantry}/d_V^2 = 20 \times 10^4/5.5^2\ \mu Gy = 6.6 \times 10^3\ \mu Gy$

The required transmission for protection $= 300/(6.6 \times 10^3 \times 1.0) = 45 \times 10^{-3}$.

The protection required would be 90 mm of lightweight concrete and at an angle of 30° would be afforded by:

$T_L = 90 \times (1 + \cos 30°)/2 = 84$ mm

So the protection afforded by the 100 mm thick roof slab is sufficient and no baffle will be required if the duct is sited within the X-ray shadow of the gantry.

Box 8.10 Final specification for CT scanning room

All walls to be protected by Code 5 lead manufactured to BSEN 12588:2006, to the height of the false ceiling.

Protection in the walls above the level of the false ceiling to be with at least Code 3 lead.

The door to the corridor to be protected by at least 1.2 mm lead (Code 3).

The control cubicle window is to be protected by 2 mm lead-equivalent glass at 120 kV.

The door to the control cubicle is to be protected by at least 1.7 mm of lead (Code 4).

The floor and ceiling require protection equivalent to 168 mm of lightweight concrete or Code 3 lead to be added to the 100 mm thick slabs.

The ventilation duct is to be sited to the left of the door, looking from the inside of the room. If this is not possible, a baffle containing Code 3 lead should be installed.

The region of highest scatter occurs at angles between 30° and 40° to the vertical to the front of the gantry, so an angle of 30° is considered when calculating the transmission requirements. However, because the level of scatter to the side of the gantry gradually increases from an angle of 20° to the scan plane, the region from 20° upwards is considered when specifying the extent of wall and door protection requirements.

CT control cubicles will always require shielded doors to protect against tertiary scatter.

The walls of any CT room will require protection with at least Code 4 lead, even if the workload is at the lower end of the normal range.

Gantry shielding should be considered in determining the location of any large penetrations.

CT simulators are used in a similar manner to conventional CT scanners, but a larger field of view is normally chosen. The amount of scatter is similar to that for other CT scanners. However, because a larger field of view is selected, the DLP displayed is likely to be calibrated for the body. The scatter factor for scans of the head when the DLP is displayed in terms of that for the body is a similar order to that for the body, so calculations of protection for CT simulators should be based on the factors for the body.

8.7 Mobile CT scanners within hospitals

Mobile head CT equipment is now available for use in intensive care. A similar methodology based on a head DLP scatter factor of 0.14 μGy (mGy cm)$^{-1}$ could be used to predict likely scatter levels from projected workloads, but the protection afforded by the gantry is likely to be less. An assessment of protection issues for a mobile CT unit has been published by Stevens et al (2009). They recommend fitting a pair of 0.5 mm lead-equivalent lead/rubber shadow shields at the side of the couch, which could be angled towards the patient to limit scatter towards the feet, and a 1.0 mm lead shield across the rear adjacent to the head to limit scatter in this direction.

Table 8.3 Summary of CR shielding calculations

Parameter	Wall A	Wall B	Door C	Vent 1C	Vent 2C	Cubicle window D	Cubicle door D	Ceiling	Ceiling scatter
Distance to wall (m)	2.5	2.0	3.2	3.2	3.2	3.5	3.5	3.6	3.5
Distance for calculation (m)	2.8	2.0	3.4	5.5	5.5	3.5	4.2	4.2	3.5
Annual air kerma at wall (μGy)	204×10^3	400×10^3	159×10^3	61×10^3	6.6×10^3	150×10^3	105×10^3	104×10^3	3.2×10^3
Occupancy (%)	50	100	20	100	100	100	100	100	100
Transmission	3.0×10^{-3}	0.75×10^{-3}	9.3×10^{-3}	5.0×10^{-3}	45×10^{-3}	2.0×10^{-3}	2.9×10^{-3}	2.9×10^{-3}	0.09
Thickness of lead (mm)	1.6	2.2	1.2	1.4	0.7	1.8	1.7	1.7	0.5
Thickness of concrete weighing 2350 kg m^{-3} (mm)	140	176	—	118	65	—	—	132	50
Thickness of concrete weighing 1840 kg m^{-3} (mm)	178	224	—	150	84	—	—	168	64